EE1010 Math for EE

Lecture 1: Sets and Mathematical Logic

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These notes summarize the discussion in class and are *not* a substitute for your own notes. They have also not been proofread, so they may have some errors (Please let the teaching team know if you encounter any).

Notation We assume basic familiarity with sets and numbers. We use the following notations

$\mathbb{N}$	set of natural numbers	
$\mathbb{Z}$	set of integers	
$\mathbb{Q}$	set of rational numbers	
$\mathbb{R}$	set of real numbers	
$\mathbb{C}$	set of complex numbers	
$\in$	belongs to	$x \in \mathbb{R}$
$\|, :$	such that (set builder)	$A = \{x \in \mathbb{R} \| 1 \leq x \leq 5\}$
$\ni$	such that (outside sets)	Find $x \ni x^2 - 2x + 1 = 0$
$\exists$	exists	$\exists x \in \mathbb{R} \ni x^2 - 2x + 1 = 0$
$\forall$	for all	$\forall x \in \mathbb{R}, x^2 - 2x + 1 \geq 0$
$\subseteq, \cup, \cap$	subset or equal, union, intersection (respectively)	

Table 1: Typical notation that you may find in math textbooks.

A mathematical statement or a *proposition* $P(x)$ returns `true` or `false` for each member x of a given set/context U. Some mathematical statements may not have a 'variable' subject x.

$P(x)$ may also be referred to as a predicate, and x as the subject. The set U is a universal set of subjects for the given context

Examples

- $\sqrt{2}$ is irrational
- $x^2 + y^2 \geq 2xy$ for any $x, y \in \mathbb{R}$.

Implication We say a statement $P(x)$ implies $Q(x)$ (and write $P(x) \implies Q(x)$) to mean that $Q(x)$ is true whenever $P(x)$ is. Implication is closely connected to sets: If $S_p = \{x | P(x) \text{ is true }\}$ and $S_q = \{x | Q(x) \text{ is true }\}$, then $P(x) \implies Q(x)$ is same as saying $S_p \subseteq S_q$. Likewise, $A \subseteq B$ can be written as

$$x \in A \implies x \in B.$$

Implication can be thought of as an *If* ... *then* ... statement.

Negation of a mathematical statement is a statement that is true whenever the original statement is false and vice versa. Negation is connected to the set-theoretic concept of a complement. The negation of P is written $NOT(P)$.

CONTRAPOSITION If $P \implies Q$ then $NOT(Q) \implies NOT(P)$. Set theoretically, this is like saying that if $A \subseteq B$ then $B^c \supseteq A^c$.

By applying this result twice, this also means that $P \implies Q$ is equivalent to $NOT(Q) \implies NOT(P)$

When $P \implies Q$ then we say that P is a *sufficient* condition for Q. Likewise, if $P \implies Q$, we say that Q is a *necessary condition* for P [1].

[1] This is because if Q does not hold (i.e. $NOT(Q)$ holds), then by the contraposition, P does not hold.

EQUIVALENCE We say that P is equivalent to Q if $P \implies Q$ and $Q \implies P$. In this case, P is a necessary and sufficient condition for Q and vice versa.

We write $P \iff Q$ or P iff Q or P if and only if Q

A MATHEMATICAL PROOF is a sequence of implications.

CONTRADICTIONS AND TAUTOLOGIES A statement which is always false (regardless of the subject) and is referred to as a *contradiction*. A statement which is always true is called a *tautology*. We have the following observations

These correspond to empty set and universal set respectively. While the terms 'contradiction' and 'tautology' are generally used for statements $P(x)$ with a subject, we retain the same terminology for all statements, even those without a subject x.

1. Every statement implies a tautology.

   Every set is a subset of the universal set

2. A contradiction implies every statement.

   The emptyset is a subset of every set

3. If a tautology implies P, then P is a tautology.

   If the universal set is contained in a set A, then A itself has to be universal

4. If P implies a contradiction, then P is a contradiction.

   If a set A is contained in an empty set, then A itself has to be an emptyset.

Statement 1. is referred to as being *trivially* true, and statement 2. is referred to as being *vacuously* true.

Statements 3. and 4. are very useful for proof techniques. A standard proof technique to prove a statement P would start with something we know to be true (i.e. a tautology) and *derive* the given statement P(i.e. 3. above). An alternate approach is to assume P is not true and obtain a contradiction (i.e. 4. above). We discussed a few examples of these.

COMPOSITE STATEMENTS are constructed by combining multiple statements; this combination is achieved via using implication ($\implies$) and its variants; or by using boolean logic (OR, AND), in addition to NOT.

The boolean OR is analogous to union of sets, AND is analogous to intersection of sets (as remarked earlier, NOT is analogous to complement of sets). As such we have the well-known *De Morgan's law*

$$(A \cup B)^c = B^c \cap A^c;$$

the logical equivalent of which is

$$NOT(P\ AND\ Q) = NOT(P)\ AND\ NOT(Q).$$

This is useful to construct negations of complicated composite statements.

Also useful to understand composite statement is to construct a *truth table* which lists the truth of the individual statements along with the final truth value.

P	Q	P OR Q
True	True	True
True	False	True
False	True	True
False	False	False

P	Q	$P \implies Q$
True	True	True
True	False	False
False	True	True
False	False	True

P	Q	$P \iff Q$
True	True	True
True	False	False
False	True	False
False	False	True

Table 2: Truth tables

Excercises What is wrong with the following proofs ?

- *To prove:* 1 is the largest integer.
  Proof: Suppose $n > 1$ is the largest integer. Then $n^2 = n.n > n.1$, contradicting that n is the largest integer. Therefore, 1 is the largest integer.

- *To prove:* $\forall x, y \in \mathbb{R}$, $x^2 + y^2 \geq 2xy$.
  Proof: We have $x^2 + y^2 \geq 2xy$ which means $x^2 - 2xy + y^2 \geq 0$, which implies $(x - y)^2 \geq 0$; which we know to be true. Hence $x^2 + y^2 \geq 2xy$.

- *To prove:* If n is an integer such that 2 divides n, then 4 divides n.
  Proof: We will prove the contrapositive of this statement and show that if n is an integer where 4 divides n, then 2 divides n. Since 4 divides n we have $n = 4r$ for some integer r. Then $n = 2(2r)$, so n is divisible by 2.

Some Questions

1. Suppose you want to prove that $P \implies Q$ is false. Which of the following approaches is correct
   - Prove that $P \implies NOT(Q)$
   - Prove that $P\ AND\ NOT(Q)$ is not false

2. We want to prove the following statement 'For any integers a, b, c if $a^2 + b^2 = c^2$, then one of a, b, c is even'. How do we go about it? Which technique do we use ?

3. Repeat the above for the statement 'For all integers n, if n is odd then so is n^2'.

4. Repeat the above for the following statement: 'For all integers m, n if mn is even and m is odd, then n is even'.

Suggested readings:

1. Chapter 3,6,7 of [2]

2. Proofwriting Checklist, CS103 Stanford [3]

3. Further reading: Propositional logic

[2] Ian Stewart and David Tall. *The foundations of mathematics*. OUP Oxford, 2015

[3] Proof checklist. `https://web.stanford.edu/class/cs103/proofwriting_checklist`. Accessed: 2024-08-6

References

[1] Proof checklist. `https://web.stanford.edu/class/cs103/proofwriting_checklist`. Accessed: 2024-08-6.

[2] Ian Stewart and David Tall. *The foundations of mathematics*. OUP Oxford, 2015.